Gaël BOYA

Embedded Software & Real-Time Systems Engineer

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France
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LANGUAGES

French
Mother tongue

English
Level C1

SOCIAL NETWORKS

Adébayo BOYA
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SKILLS

Embedded Software
C/C++, Python, Memory management, Drivers

OS & RTOS
FreeRTOS, Bare-metal, Embedded Linux

Microcontrollers
STM32 (Cortex-M3/M4), ESP32-S3, Arduino

Protocols
UART, SPI, I2C, AXI DMA, HTTP, MQTT

Tools
STM32CubeIDE, PlatformIO, VS Code, GDB, Make

Versionning
Git, SVN

SOFT SKILLS

Attention to detail

Analytical mindset

Autonomy & initiative

Team collaboration

Technical documentation

Embedded Systems Engineer (Grenoble-INP Phelma & INPT), specializing in low-level software development (C/C++) and real-time operating systems (RTOS). My strong understanding of the hardware layer (microcontrollers, RISC-V architectures) allows me to develop highly optimized firmware. I am seeking an embedded developer position to tackle technical challenges in IoT, automotive, or robotics.

EXPERIENCES

Embedded System & IoT Trainer (Freelance)

From March 2026 to April 2026 SUPINFO Lyon

Delivered a 35h intensive course for engineering students : microcontrollers, sensors, communication protocols and logic circuits.

SoC Integration Intern Engineer

From February 2024 to August 2024 Silicon Mobility Valbonne, France

- RTL design & top-level integration of a reference SoC (FPCU): architecture decisions, IP block integration, exhaustive RTL simulation and functional verification.
- Improved internal SoC integration tooling and authored technical documentation (user guide), streamlining future project delivery timelines.

Engineering Intern in Architecture & Security

From June 2023 to August 2023 Institut National des Postes et Télécommunications Rabat, Morocco

- Microarchitecture modelling of CPU/SoC designs using gem5 simulator; study, simulation and countermeasure implementation against side-channel attacks (Spectre/Meltdown, Flush+Reload).

EDUCATION

Engineering Degree — Integrated Electronic Systems

From September 2023 to April 2025 Grenoble-INP PHELMA Grenoble, France

Engineering Degree — Embedded Systems and Digital Services

From October 2021 to July 2025 Institut National des Postes et Télécommuniations (INPT) Rabat, Morocco

PROJECTS

STM32 Bare-Metal OLED Driver (SSD1331)

February 2026 Grenoble-INP Phelma Grenoble, France

- Full bare-metal SPI2 driver for Pmod OLEDrgb on STM32 Nucleo-F103RB (Cortex-M3) : custom init sequence, VCCEN/PMODEN control, RGB565 graphics primitives, 5x7 ASCII bitmap font engines
- Zero external librairies : *printf* retargeted to USART2 via ST-LINK for serial debugging.

Edge AI Surveillance Node — ESP32-S3

February 2026 Personal Grenoble, France

- FreeRTOS multi-task architecture on ESP32-S3 : independent tasks for embedded HTTP server and hardware heartbeat; MJPEG stream optimised with external PSRAM and DMA double-buffering for smooth Wi-Fi video.
- Decoupled Python/OpenCV node consuming the HTTP server stream for real-time Haar Cascade face detection.

Real-Time Control and Command — NXP Autonomous Vehicle

From October 2024 to January 2025 Grenoble-INP Phelma Grenoble, France

- Implemented FreeRTOS control loops : wheel speed servo and steering with empirical gain tunning.
- Real-time camera pipeline : image capture → gaussian filter → edge detection → lane-tracking servo loop.